

tf.compat.v1.nn.sufficient_statistics Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/compat/v1/nn/sufficient_statistics

Calculate the sufficient statistics for the mean and variance of x .

Function Signatures

```
tf.compat.v1.nn.sufficient_statistics( x, axes, shift=None,
keep_dims=None, name=None, keepdims=None )
sufficient_statistics(t, [1])
(,
```

Code Examples

```
tf.compat.v1.nn.sufficient_statistics(
x, axes, shift=None, keep_dims=None, name=None, keepdims=None
)
```

```
tf.compat.v1.nn.sufficient_statistics(
x, axes, shift=None, keep_dims=None, name=None, keepdims=None
)
```

```
t = [[1, 2, 3], [4, 5, 6]]
sufficient_statistics(t, [1])
(, , , None)
sufficient_statistics(t, [-1])
(, , , None)
```

```
t = [[1, 2, 3], [4, 5, 6]]
```

```
sufficient_statistics(t, [1])
```

```
(,
```

```
dtype=int32, numpy=array([ 6, 15], dtype=int32)>,
```

```
dtype=int32, numpy=array([14, 77], dtype=int32)>, None)
```

Documentation

Calculate the sufficient statistics for the mean and variance of x .

These sufficient statistics are computed using the one pass algorithm on an input that's optionally shifted. See: https://en.wikipedia.org/wiki/Algorithms_for_calculating_variance#Computing_shifted_data

For example:

Returns

- the count (number of elements to average over).
- the (possibly shifted) sum of the elements in the array.
- the (possibly shifted) sum of squares of the elements in the array.
- the shift by which the mean must be corrected or None if shift is None.

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